

Robust Controller Synthesis

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Abstract—

Index Terms—PDEs, Robust Control, LMIs;

I. INTRODUCTION

This paper uses PN PI operators inspired by the works [2], [5], [13]. Then it IQC extends the results from [1], [4], [11]. [3] also covers robust control using dual iq

II. PRELIMINARIES

A. Notation

$L_2^n[a, b]$ denotes the set of $\mathbb{R}^n$ -valued, Lebesgue square-integrable equivalence classes of functions on the spatial domain $[a, b] \subset \mathbb{R}$. The space $\mathbb{R}L_2^n[a, b]$ denotes the Cartesian product $\mathbb{R}^{n_1} \times L_2^{n_2}[a, b]$, where $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$, with inner product

$$\left\langle \begin{bmatrix} x_1 \\ \mathbf{x}_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ \mathbf{y}_2 \end{bmatrix} \right\rangle_{\mathbb{R}L_2} := x_1^\top y_1 + \langle \mathbf{x}_2, \mathbf{y}_2 \rangle_{L_2}.$$

When clear from context, we occasionally omit the subscript and simply write $\langle \cdot, \cdot \rangle := \langle \cdot, \cdot \rangle_{\mathbb{R}L_2}$. Since the only spatial domain considered here is $[a, b]$, we typically omit the domain and write L_2^n or $\mathbb{R}L_2^n$, further omitting dimensions when clear from context. For a normed space X and Banach space Y , $\mathcal{L}(X, Y)$ denotes the Banach space of bounded linear operators from X to Y with induced norm

$$\|\mathcal{P}\|_{\mathcal{L}(X, Y)} = \sup_{\|\mathbf{x}\|=1} \|\mathcal{P}\mathbf{x}\|_Y,$$

and $\mathcal{L}(X) := \mathcal{L}(X, X)$. Our notational convention is to write functions in **bold** (e.g., $\mathbf{x}$, indicating $\mathbf{x} \in \mathbb{R}L_2$) and operators in calligraphic font (e.g., $\mathcal{P} \in \mathcal{L}(\mathbb{R}L_2)$). For a Hilbert space X and operator $\mathcal{A} \in \mathcal{L}(X)$, the adjoint $\mathcal{A}^*$ is defined by

$$\langle \mathbf{x}, \mathcal{A}\mathbf{y} \rangle_X = \langle \mathcal{A}^*\mathbf{x}, \mathbf{y} \rangle_X, \quad \forall \mathbf{x}, \mathbf{y} \in X.$$

The space $L_2^n[0, \infty)$ denotes $\mathbb{R}^n$ -valued square-integrable signals defined on the temporal domain $[0, \infty)$, so that $x(t) \in \mathbb{R}^n$. Similarly, $\mathbb{R}L_2^n[0, \infty)$ denotes $\mathbb{R}L_2$ -valued square-integrable temporal signals. For suitably differentiable $\mathbf{x} \in \mathbb{R}L_2^n[0, \infty)$, we write $\dot{\mathbf{x}} := \frac{\partial \mathbf{x}}{\partial t}$. For a Hilbert space $\mathbf{H}$, we denote by $\mathbf{L}_{\mathbf{H}}$ the space of square-integrable functions $u : [0, \infty) \rightarrow \mathbf{H}$ with inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{L}_{\mathbf{H}}} = \int_0^\infty \langle \mathbf{u}(t), \mathbf{v}(t) \rangle_{\mathbf{H}} dt.$$

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We define the extended space $\mathbf{L}_{e, \mathbf{H}}$ to consist of functions $u : [0, \infty) \rightarrow \mathbf{H}$ such that

$$\int_0^T \|\mathbf{u}(t)\|_{\mathbf{H}}^2 dt < \infty \quad \forall T \geq 0.$$

We use $\mathbf{L}_{e, [a, b]}$ and $\mathbf{L}_{[a, b]}$ to denote $\mathbf{L}_{e, \mathbf{H}}$ and $\mathbf{L}_{\mathbf{H}}$ with $\mathbf{H} = \mathbb{R}L_2[a, b]$. For any Hilbert space $\mathbf{H}$, the truncation operator $p_T : \mathbf{L}_{e, \mathbf{H}} \rightarrow \mathbf{L}_{e, \mathbf{H}}$ is defined for $\mathbf{y} \in \mathbf{L}_{e, \mathbf{H}}$ by

$$(p_T \mathbf{y})(t) = \begin{cases} \mathbf{y}(t), & 0 \leq t \leq T, \\ 0, & \text{otherwise.} \end{cases}$$

such that $(p_T \mathbf{y})(t) \in \mathbf{L}_{\mathbf{H}}$ for all $T \geq 0$.

Definition 1. Let $\mathbf{H}$ and $\mathbf{G}$ be Hilbert spaces. For all $T \geq 0$, an operator $G : \mathbf{L}_{\mathbf{H}} \rightarrow \mathbf{L}_{\mathbf{G}}$ is:

- **Causal** if $p_T G = p_T G p_T$.
- **Bounded on $\mathbf{L}_{\mathbf{H}}$** if $\|G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|\mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{\mathbf{H}}$.
- **Bounded on $\mathbf{L}_{e, \mathbf{H}}$** if $\|p_T G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|p_T \mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{e, \mathbf{H}}$.

Finally, we will note some operations. For a given $\mathcal{K} \in \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, we define the operator $\mathcal{M}_{\mathcal{K}} : \mathbf{L}_{e, [a, b]}^{\mathbf{n}} \rightarrow \mathbf{L}_{e, [a, b]}^{\mathbf{m}}$ to compactly represent a multiplication operator. For $\mathbf{u} = [u_1 \ \cdots \ u_N]^\top$ with $\mathbf{u} \in \mathbb{R}L_2$, its action at time t is given by the block structure of $\mathcal{K}_{ij}$:

$$(\mathcal{M}_{\mathcal{K}} \mathbf{u})(t) := \begin{bmatrix} \mathcal{K}_{11} & \cdots & \mathcal{K}_{1n} \\ \vdots & \ddots & \vdots \\ \mathcal{K}_{m1} & \cdots & \mathcal{K}_{mn} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{bmatrix} (t).$$

B. Partial Integral Equations (PIEs)

Partial Integral operators, also known as PI operators, are a class of bounded linear integral operators that are defined jointly on a vector space and Hilbert space.

Definition 2. We say $\mathcal{P} \in \Pi_4^{\mathbf{m} \times \mathbf{n}} \subset \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, denoted by $\Pi \begin{bmatrix} P & Q_1 \\ Q_2 & \{R\} \end{bmatrix}$, if there exist a matrix P and matrix polynomials Q_1, Q_2, R_0, R_1 , and R_2 (of compatible dimension) such that,

$$\left[\left(\Pi \begin{bmatrix} P & Q_1 \\ Q_2 & \{R\} \end{bmatrix} \right) \begin{bmatrix} x \\ \mathbf{x} \end{bmatrix} \right] (s) := \begin{bmatrix} Px + \int_a^b Q_1(s) \mathbf{x}(s) ds \\ Q_2(s)x + \mathcal{R}\mathbf{x}(s) \end{bmatrix},$$

$$(\mathcal{R}\mathbf{x})(s) = R_0(s)\mathbf{x}(s) + \int_a^s R_1(s, \theta)\mathbf{x}(\theta) d\theta + \int_s^b R_2(s, \theta)\mathbf{x}(\theta) d\theta.$$

The vector space of PI operators, given compatible dimensions, are closed under composition, addition, adjoint and

concatenation – implying that Π_4 is a composition algebra [10].

Definition 3. (LPIs) For given PI operators $\{\mathcal{E}_{i,j}, \mathcal{F}_{i,j}, \mathcal{G}_i\}$ and a convex linear functional $\mathcal{L}(\cdot)$, a linear PI inequality is a convex optimization of the following form

$$\min_{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}} \mathcal{L}(\{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}\})$$

$$\text{s.t. } \sum_{j=1}^K \mathcal{E}_{ij}^* \Pi \begin{bmatrix} P_i & Q_{1i} \\ Q_{2i} & R_i \end{bmatrix} \mathcal{F}_{ij} + \mathcal{G}_i \succeq 0$$

Only LMIs are required to solve an optimization problem involving LPIs. These definitions are based on [7]

Definition 4 (PIE System). Given the PIE system $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ with PI operators $\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}$, and $\mathcal{D}$ of compatible dimensions, and signals $w(t) \in \mathbb{R}L_2^{\mathbf{n}_w}[a, b]$, we say $\mathbf{x}(t) \in \mathbb{R}L_2[a, b]^{\mathbf{n}_x}$ and $z(t) \in \mathbb{R}L_2^{\mathbf{n}_z}[a, b]$ satisfy the PIE with initial condition $\mathbf{x}_0$ if

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x})(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix}$$

for all $t \geq 0$ and $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$. Additionally, for signals $\underline{w}(t) \in \mathbb{R}L_2[a, b]^{\mathbf{n}_z}$ of compatible dimensions, we say $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2[a, b]^{\mathbf{n}_x}$ and $\underline{z}(t) \in \mathbb{R}L_2[a, b]^{\mathbf{n}_w}$ satisfy the dual PIE, $G^\top$, with initial condition $\underline{\mathbf{x}}_0$ if

$$\begin{bmatrix} \mathcal{T}^*\underline{\dot{\mathbf{x}}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix},$$

for all $t \geq 0$ and $\mathcal{T}^*\underline{\mathbf{x}}(0) = \mathcal{T}^*\underline{\mathbf{x}}_0$.

Definition 5 (Bounded Internally Stable PIE). Let G be a PIE system as in Definition 4. We say G is internally stable and bounded with decay rate $\alpha > 0$, $M > 0$, and L_2 -gain $\gamma > 0$ if

- The homogeneous part $\partial_t(\mathcal{T}\mathbf{x})(t) = \mathcal{A}\mathbf{x}(t)$, with $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$ is exponentially stable such that, for every admissible initial condition $\mathbf{x}_0 \in \mathbb{R}L_2$

$$\|\mathcal{T}\mathbf{x}(t)\|_{\mathbf{L}_{[a,b]}} \leq M\|\mathcal{T}\mathbf{x}_0\|_{\mathbf{L}_{[a,b]}} e^{-\alpha t}, \quad \forall t \geq 0.$$

- With zero initial condition $\mathbf{x}(0) = 0$, then for any input w and output z ,

$$\|p_T z\|_{\mathbf{L}_{[a,b]}} \leq \gamma \|p_T w\|_{\mathbf{L}_{[a,b]}}, \quad \forall T \geq 0, \quad \forall w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}.$$

Then $G \in \mathcal{G}_2^{\mathbf{n}_z \times \mathbf{n}_w}$. Clearly, $\mathcal{R}H_\infty \subset \mathcal{G}_2$ [15]. Also $G^\top \in \mathcal{G}_2$ with the same decay rate, $\mathbf{L}_{[a,b]}$ -gain, and M [7].

Definition 6 (Linear Fractional Transformation (LFT)). Let G be a causal system or operator with a specified 2×2 block partition, and let H be compatible with either two ports or one port. Write

$$\begin{bmatrix} y_{11} \\ y_{12} \end{bmatrix} = G \begin{bmatrix} u_{11} \\ u_{12} \end{bmatrix}.$$

The Redheffer star products [6] are the following well-posed interconnections.

- If H has two ports,

$$\begin{bmatrix} y_{21} \\ y_{22} \end{bmatrix} = H \begin{bmatrix} u_{21} \\ u_{22} \end{bmatrix},$$

then

$$G \star H : \begin{bmatrix} u_{12} \\ u_{21} \end{bmatrix} \mapsto \begin{bmatrix} y_{12} \\ y_{21} \end{bmatrix}, \quad u_{11} = y_{22}, \quad u_{22} = y_{11},$$

$$H \star G : \begin{bmatrix} u_{11} \\ u_{22} \end{bmatrix} \mapsto \begin{bmatrix} y_{11} \\ y_{22} \end{bmatrix}, \quad u_{21} = y_{12}, \quad u_{12} = y_{21}.$$

- If H has one port, $y_H = H u_H$, then

$$G \star H : u_{12} \mapsto y_{12}, \quad u_{11} = y_H, \quad u_H = y_{11},$$

$$H \star G : u_{11} \mapsto y_{11}, \quad u_H = y_{12}, \quad u_{12} = y_H.$$

C. PIETOOLS

PIETOOLS is a MATLAB toolbox used for declaration, manipulation and solving LPIs [9]. For learning how to use PIETOOLS and its functionalities visit PIETOOLS homepage.

III. IQC-BASED FRAMEWORK FOR ROBUST STABILITY AND PERFORMANCE ANALYSIS

This section develops a comprehensive framework for analyzing stability and performance of interconnected systems in a Hilbert space, utilizing IQCs as the primary analytical tool. We begin by establishing the mathematical foundation for system interconnections and formalizing the key concepts of stability and performance. The IQC methodology is then systematically developed to characterize system uncertainties and derive rigorous conditions that guarantee robust stability and performance.

A. Description of uncertain systems in a Hilbert-space

We consider operators that map external inputs to external outputs through a feedback interconnection between a linear (infinite dimensional) operator P and Δ . This interconnection structure is formally defined as follows

Definition 7 (Generalized Plant). Given operators $P_\Delta : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$, $P_{\Delta P} : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_P}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$, $P_{P\Delta} : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_P}}$, $P_P : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_P}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_P}}$, and $\Delta : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_\Delta}}$, define

$$\begin{bmatrix} z_\Delta \\ z_P \end{bmatrix} = P \begin{bmatrix} w_\Delta \\ w_P \end{bmatrix} = \begin{bmatrix} P_\Delta & P_{\Delta P} \\ P_{P\Delta} & P_P \end{bmatrix} \begin{bmatrix} w_\Delta \\ w_P \end{bmatrix}, \quad w_\Delta = \Delta z_\Delta. \quad (1)$$

for input $w_P \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_P}}$ and output $z_P \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_P}}$, there exist internal signals $z_\Delta \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$ and $w_\Delta \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_\Delta}}$ satisfying the interconnected system $z_P = (P \star \Delta)w_P$.

The interconnection structure described above captures the feedback interaction between the nominal system P and the uncertainty Δ , where w_P and z_P represent external inputs and performance outputs, respectively, while w_Δ and z_Δ form the internal feedback loop through the uncertainty. Before analyzing stability and performance, we must first ensure the well-posedness of the interconnection, such that solutions exist, are unique, and depend causally on inputs.

Definition 8. We say that the uncertain interconnection $P \star \Delta$ is *well-posed* if:

- **Existence and Uniqueness:** For any $w \in \mathbf{L}_{e,[a,b]}$, there exist unique $z \in \mathbf{L}_{e,[a,b]}$, $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and $w_\Delta \in \mathbf{L}_{e,[a,b]}$ such that, z , w , z_Δ , w_Δ satisfy the interconnection.
- **Causality:** If z_Δ , w_Δ and $\hat{z}_\Delta$, $\hat{w}_\Delta$ satisfy the interconnection, then $p_T(z_\Delta - \hat{z}_\Delta) = 0$ and $p_T(w_\Delta - \hat{w}_\Delta) = 0$, for all $T \geq 0$.

Having established the class of uncertain systems and their well-posedness, we now develop a framework to guarantee robust stability and performance for the class of systems defined in Definition 7.

B. Robust stability and performance

Traditional control system design often treats stability and performance shaping as separate objectives with distinct constraints. However, Integral Quadratic Constraints (IQCs) provide a unified framework where these trade-offs can be captured within a single constraint [11]. We now define what it means for the system to remain stable despite uncertainties. Robust stability requires that the system is well-posed for all admissible uncertainties and that the input-output map remains bounded. This involves removal of the performance channels in (1), and introducing auxiliary signals z_0 and w_0 .

Definition 9. *The interconnection*

$$z_\Delta = P_\Delta w_\Delta + z_0, \quad w_\Delta = \Delta z_\Delta + w_0,$$

with $z_0, w_0 \in \mathbf{L}_{e,[a,b]}$, is said to be **robustly stable** if:

- 1) The uncertain system is well-posed for all $\Delta \in \mathbf{\Delta}$.
- 2) The map from $\begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$ to $\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ is bounded on $\mathbf{L}_{e,[a,b]}$ for all $\Delta \in \mathbf{\Delta}$.

Within the IQC framework, the stability of an interconnection $P \star \Delta$ is certified by verifying that the uncertainty operator Δ satisfies a quadratic constraint defined by an appropriate multiplier. This leads to the following formal definition

Definition 10. An uncertainty operator $\Delta : \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$ is said to satisfy the hard uncertainty IQC $(\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta})$ defined by the dynamic operator $\Psi_\Delta : \mathbf{L}_{e,[a,b]}^{n_{z_\Delta} + n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta} + n_{w_\Delta}}$ multiplication operator $\mathcal{M}_{\mathcal{V}_\Delta}$, parameterized by $\mathcal{V}_\Delta = \mathcal{V}_\Delta^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ if the inequality holds for all $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and all $T \geq 0$,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0.$$

Moreover, let $\mathcal{V}_\Delta \{ \mathcal{Q}_\Delta, \mathcal{S}_\Delta, \mathcal{R}_\Delta \}$ be parameterized as,

$$\mathcal{V}_\Delta = \begin{bmatrix} \mathcal{Q}_\Delta & \mathcal{S}_\Delta \\ \mathcal{S}_\Delta^* & \mathcal{R}_\Delta \end{bmatrix}$$

Beyond stability, we are interested in ensuring that the system meets certain performance specifications in the presence of uncertainty. Robust performance extends the notion of robust stability by additionally requiring that a performance metric is satisfied.

Definition 11. *Robust performance* is achieved if the interconnection $P \star \Delta$ is **robustly stable** for some hard performance IQC $(\Psi_p, \mathcal{M}_{\mathcal{V}_p})$ defined by the dynamic operator

$\Psi_p : \mathbf{L}_{e,[a,b]}^{n_{z_p} + n_{w_p}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_p} + n_{w_p}}$ and multiplication operator $\mathcal{M}_{\mathcal{V}_p}$, parameterized by $\mathcal{V}_p = \mathcal{V}_p^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ with

$$\left\langle p_T \Psi_p \begin{bmatrix} v \\ 0 \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} v \\ 0 \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0.$$

for all $v \in \mathbf{L}_{e,[a,b]}^{n_{z_p}}$. Then there exists a sufficiently small $\epsilon > 0$ such that, for all such w_p and all $T \geq 0$,

$$\left\langle p_T \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_p\|_{\mathbf{L}_{[a,b]}}^2.$$

Moreover, let $\mathcal{V}_p \{ \mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p \}$ be parameterized as,

$$\mathcal{V}_p = \begin{bmatrix} \mathcal{Q}_p & \mathcal{S}_p \\ \mathcal{S}_p^* & \mathcal{R}_p \end{bmatrix}$$

In this framework, the multipliers $\mathcal{M}_{\mathcal{V}_\Delta}$ and $\mathcal{M}_{\mathcal{V}_p}$ serve distinct but complementary roles. The hard uncertainty IQC $(\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta})$ characterizes the behavior of uncertainties, nonlinearities, and other perturbations captured by the operator Δ , ensuring that the IQC is satisfied for the class of admissible disturbances. In contrast, the hard performance IQC $(\Psi_p, \mathcal{M}_{\mathcal{V}_p})$ encodes performance requirements for the nominal system, such as L_2 -gain, passivity conditions, or other desired input-output properties. Together, these multipliers enable simultaneous verification of robust stability against perturbations and achievement of specified performance objectives.

C. Theorem for joint robust stability and performance

We now present the theoretical result that provides conditions for simultaneous robust stability and performance verification. For brevity we will refer to the hard IQC as,

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2$$

Let Ψ_Δ and Ψ_p have a 2×2 block partition and define $\Psi_{ij} := \text{diag}((\Psi_\Delta)_{ij}, (\Psi_p)_{ij})$ for $i, j \in \{1, 2\}$. Let $w^\top = [w_\Delta^\top \ w_p^\top]$ and $z^\top = [z_\Delta^\top \ z_p^\top]$. Moreover, define $\mathcal{Q} = \text{diag}(\mathcal{Q}_\Delta, \mathcal{Q}_p)$, $\mathcal{S} = \text{diag}(\mathcal{S}_\Delta, \mathcal{S}_p)$, and $\mathcal{R} = \text{diag}(\mathcal{R}_\Delta, \mathcal{R}_p)$, then $\mathcal{V} \{ \mathcal{Q}, \mathcal{S}, \mathcal{R} \}$.

Theorem 1 (Robust Stability and Performance). *Let P , $\Delta \in \mathbf{\Delta}$ be given, and there exists $\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta}$, such that,*

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \mathbf{\Delta}$
- The IQC in Definition 10 is satisfied for all $\Delta \in \mathbf{\Delta}$.

*Then the interconnection $P \star \Delta$ is **robustly stable** and **robust performance** is achieved with respect to some Ψ_p and $\mathcal{M}_{\mathcal{V}_p}$, in Definition 11, with $\mathcal{Q}_p \succeq 0$, if,*

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2 \quad (2)$$

for all $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$, $w_p \in \mathbf{L}_{e,[a,b]}^{n_{w_p}}$ and $T \geq 0$.

Proof. See Appendix A.1 ■

Theorem 1 establishes a framework for deriving Linear PI Inequalities (LPIs), which are used in assessing the robust performance of a system.

IV. IQC DUALITY

In this work we assume that the generalized plant $P \in \mathcal{G}_2^{\mathbf{n}_z \times \mathbf{n}_w}$ is a PIE system, that admits a state-space representation. This work builds on the duality results in [7] which shows that the state-feedback controller results in a convex optimization problem. We find that this duality result can be extended to the case of dynamic filters, where we want to know the properties of the dual dynamic filter. This is particularly useful when examining the effect of uncertainties on the system when performing state-feedback controller synthesis, when using IQCs. The works from Seiler and Venkataraman [8], [13] show that, after J -spectral factorization, the dual of a dynamic filter can be found using the Potapov-Ginzburg transform. This section extends these results to PIE systems, and shows that the dual of a PIE system with a dynamic filter can be found using the Potapov-Ginzburg transform, identical to the work from Venkataraman.

A. Duality under Dynamical Filtering

We begin by restating the main theorem from [7].

Theorem 2 (Dual L_2 -gain [7, Thm. 9]). *For a PIE system with corresponding PI operators $P\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\} \in \mathcal{G}_2$, the following statements are equivalent.*

- 1) *For any $w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}$, and $\mathbf{x}(0) = 0$, P satisfies $\|p_T z\| \leq \|p_T w\|$.*
- 2) *For any $\underline{w} \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}$, and $\underline{\mathbf{x}}(0) = 0$, $P^\top$ satisfies $\|p_T \underline{z}\| \leq \|p_T \underline{w}\|$.*

Relating an IQC to this gain duality requires two transformations. First, write $\begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix} := \Theta \begin{bmatrix} z \\ w \end{bmatrix}$. A J -spectral factorization gives, for every $T \geq 0$,

$$\begin{aligned} & \left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_V \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \\ &= \|p_T \tilde{z}\|_{\mathbf{L}_{[a,b]}}^2 - \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2. \end{aligned}$$

Thus the IQC quadratic form becomes the difference between the filtered output and input signals. Second, when $z = Pw$, the Potapov-Ginzburg transform produces an invertible change of coordinates Γ such that

$$\Theta \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} P \star (\Theta^\dagger R) \\ I \end{bmatrix} \Gamma w.$$

The lower block gives $\tilde{w} = \Gamma w$, while the upper block gives $\tilde{z} = (P \star (\Theta^\dagger R)) \tilde{w}$. Consequently, the IQC inequality becomes an L_2 -gain inequality for $P \star (\Theta^\dagger R)$, to which Theorem 2 applies. Define the channel swapping operator, so that

$$R \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} w \\ z \end{bmatrix}, \quad R \begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} z \\ w \end{bmatrix}.$$

Definition 12 (J -Spectral Factorization and Dual Filter [13]). *Let $(\Psi, \mathcal{M}_V)$ be a hard IQC, let*

$$J_{\mathbf{n}_z, \mathbf{n}_w} := \begin{bmatrix} I_{\mathbf{n}_z} & 0 \\ 0 & -I_{\mathbf{n}_w} \end{bmatrix},$$

and let $\Theta, \Theta^{-1} \in \mathcal{G}_2$. The pair $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ is a J -spectral factorization of $(\Psi, \mathcal{M}_V)$ if,

$$\begin{aligned} & \left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_V \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \\ &= \left\langle p_T \Theta \begin{bmatrix} z \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}. \end{aligned}$$

Define the matrix $\hat{J} := \begin{bmatrix} 0_{\mathbf{n}_z \times \mathbf{n}_w} & I_{\mathbf{n}_z} \\ -I_{\mathbf{n}_w} & 0_{\mathbf{n}_w \times \mathbf{n}_z} \end{bmatrix}$. The dual filter associated with Θ is defined by

$$\mathbf{D}(\Theta) := (\hat{J}^* \Theta^\top \hat{J})^{-1}.$$

Note $\mathbf{D}(\Theta) \in \mathcal{G}_2$.

Finding a J -spectral factorization is the first step in applying the duality theory. For the construction of a J -spectral factorization, we partition

$$\Theta = \begin{bmatrix} \Theta_z & \Theta_{zw} \\ \Theta_{wz} & \Theta_w \end{bmatrix}.$$

The associated primal and dual filtered signals are

$$\Theta \begin{bmatrix} z(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix}, \quad \mathbf{D}(\Theta) \begin{bmatrix} \underline{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix}.$$

The J -spectral factorization for the primal IQC leads to the state-space realization

$$\begin{bmatrix} \mathcal{T}_\Theta \dot{\mathbf{x}}_\Theta(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Theta & \mathcal{B}_\Theta^z & \mathcal{B}_\Theta^w \\ \mathcal{C}_\Theta^z & \mathcal{D}_\Theta^z & \mathcal{D}_\Theta^{zw} \\ \mathcal{C}_\Theta^w & \mathcal{D}_\Theta^{wz} & \mathcal{D}_\Theta^w \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ z(t) \\ w(t) \end{bmatrix},$$

where the entries are PI operators of appropriate dimension. The second step is the required graph rearrangement, given by the Potapov-Ginzburg transform. We use the following construction from [13].

Definition 13. (Potapov-Ginzburg Transform) *Given $\Theta : \begin{bmatrix} z \\ w \end{bmatrix} \rightarrow \begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix}$ as in Definition 12, the Potapov-Ginzburg transform denoted by $\Theta^\dagger : \begin{bmatrix} z \\ \tilde{w} \end{bmatrix} \rightarrow \begin{bmatrix} \tilde{z} \\ w \end{bmatrix}$ is defined by*

$$\Theta^\dagger = \begin{bmatrix} \Theta_z - \Theta_{zw} \Theta_w^{-1} \Theta_{wz} & \Theta_{zw} \Theta_w^{-1} \\ -\Theta_w^{-1} \Theta_{wz} & \Theta_w^{-1} \end{bmatrix},$$

The transform identifies the filtered graph of P with the graph of $P \star (\Theta^\dagger R)$, under an invertible change of coordinates. The next lemma establishes this relation for both the primal and dual interconnections.

Lemma 1. (Potapov-Ginzburg Transform Equivalence) *Let $\Theta, \mathbf{D}(\Theta) \in \mathcal{G}_2$ as in Definition 12. Moreover, let $(P \star (\Theta^\dagger R)) \in \mathcal{G}_2$ and $(P^\top \star (R(\Theta^\dagger)^\top)) \in \mathcal{G}_2$ be well-posed PIE systems, such that*

$$(\mathcal{D}_\Theta^{wz} \mathcal{D} + \mathcal{D}_\Theta^w)^{-1},$$

exist and are closed under inversion. Then, there exists invertible coordinate transformations $\mathcal{M}_\Gamma, \mathcal{M}_{\Gamma^{-1}} \in \Pi_4$ on the primal side and $\mathcal{M}_{\bar{\Gamma}}, \mathcal{M}_{\bar{\Gamma}^{-1}} \in \Pi_4$ on the dual side such that

$$\bullet \quad \Theta \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} P \star (\Theta^\dagger R) \\ I \end{bmatrix} \mathcal{M}_\Gamma w \text{ for all } w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}.$$

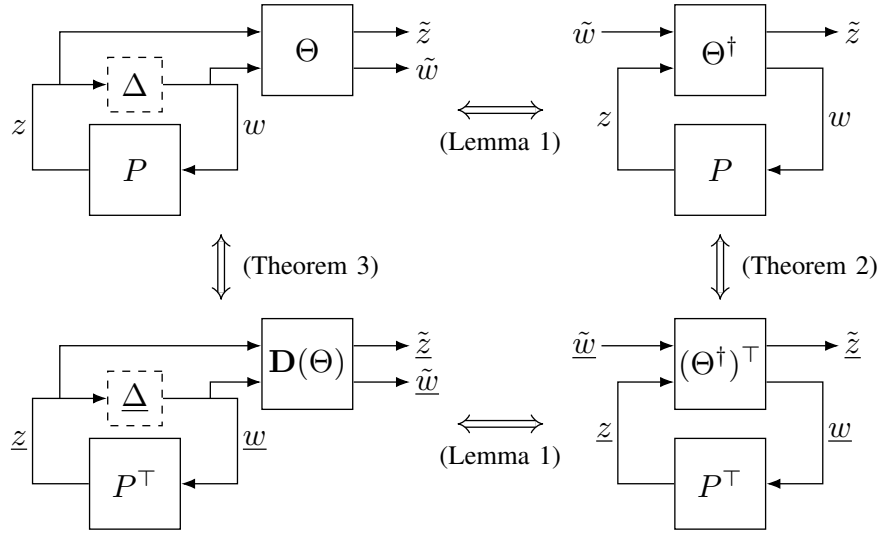


Fig. 1. Commutative diagram for primal and dual dynamic filtering under the Potapov–Ginzburg transform. [13]

$$\bullet \mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} P^\top \star (R(\Theta^\dagger)^\top) \\ I \end{bmatrix} \mathcal{M}_\Gamma \underline{w} \text{ for all } \underline{w} \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}.$$

Proof. See Appendix A.2. ■

B. Application on Dual IQC

Lemma 1 places the primal and dual filtered graphs in the form required by Theorem 2. Combining these results shows that strict feasibility can be tested on either the primal system or its dual.

Theorem 3 (Primal–Dual IQC Feasibility). *Let P be a PIE realization of Definition 7, and let $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ be a J -spectral factorization of the primal hard IQC. Let sufficiently small $\epsilon > 0$ and $\underline{\epsilon} > 0$ denote the primal and dual feasibility margins. Then the primal feasibility condition*

$$\left\langle p_T \Theta \begin{bmatrix} P \\ I \end{bmatrix} w, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} P \\ I \end{bmatrix} w \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2,$$

for all $w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}$, and $T \geq 0$, holds if and only if the dual feasibility condition

$$\left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\underline{\epsilon} \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2,$$

for all $\underline{w} \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}$, and $T \geq 0$.

Proof. See Appendix A.3. ■

The preceding theorem transfers feasibility between the dual and primal systems. To turn either description into a computational certificate for robust stability and performance, we combine it with the PIE KYP condition.

Theorem 4. *Let the realization of P satisfy Definition 7. Let $\Delta \in \Delta$ and there exist **either**, a primal hard IQC $(\Psi, \mathcal{M}_\Psi)$, with $\mathcal{V} \in \Pi_4$, or a dual hard IQC $(\underline{\Psi}, \mathcal{M}_{\underline{\Psi}})$, with $\mathcal{V} \in \Pi_4$, such that,*

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \Delta$.

- The IQC in Definition 10 is satisfied for all $\Delta \in \Delta$.
- The primal hard IQC $(\Psi, \mathcal{M}_\Psi)$ has a $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ factorization.
- The dual hard IQC $(\underline{\Psi}, \mathcal{M}_{\underline{\Psi}})$ has a $(\mathbf{D}(\Theta), J_{\mathbf{n}_w, \mathbf{n}_z})$ factorization.

Suppose that **either** of the two statements hold for some $\mathcal{Q} \succeq 0$ and $\mathcal{P} = \mathcal{P}^* \succeq 0 \in \Pi_4$.

$$\bullet \begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} \mathcal{V} \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0$$

$$\bullet \begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \underline{\epsilon} I \end{bmatrix} + \begin{bmatrix} \underline{\mathcal{C}}^* \\ \underline{\mathcal{D}}^* \end{bmatrix} \mathcal{V} \begin{bmatrix} \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \preceq 0 \quad (3)$$

Then for any $w, \underline{w} \in \mathbf{L}_{e,[a,b]}$, satisfies either

$$\Psi \begin{bmatrix} P \\ I \end{bmatrix} w := \left\{ \begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \right\}$$

or

$$\underline{\Psi} \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T} \dot{\underline{\mathbf{x}}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix} \right\} \quad (4)$$

Then the interconnection $P \star \Delta$ with $\mathbf{x}(0) = 0$ is robustly stable and robust performance is achieved.

Proof. See Appendix A.4. ■

Remark 1. Note that we do not require information about the dual uncertainty $\underline{\Delta}$, only the primal uncertainty Δ .

V. APPLICATION OF DUAL IQC

Theorem 4 provides a robust-performance certificate for PIE systems in terms of the dual IQC. For analysis, this merely reformulates the primal certificate, since primal–dual equivalence preserves feasibility. The benefit of the dual formulation therefore appears in controller synthesis. Both controller constructions below use the same synthesis LPI, obtained by applying (3) to the corresponding closed-loop PIE. The choice of dual uncertainty filter changes only the realization of the controller recovered from this LPI. A related application of IQC duality to convex feedforward controller synthesis is given by Venkataraman and Seiler [13].

A. Primal and Dual Systems

Let P , with zero initial condition $\mathbf{x}(0) = 0$, have input w , control input u , output z , and PIE realization

$$\begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P(t) \\ z(t) \\ y_P(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P & \mathcal{B}_P & \mathcal{B}_u \\ \mathcal{C}_P & \mathcal{D}_P & \mathcal{D}_{zu} \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ w(t) \\ u(t) \end{bmatrix}.$$

and its dual representation

$$\begin{bmatrix} \mathcal{T}_P^* \dot{\mathbf{x}}_P(t) \\ \underline{z}(t) \\ \underline{y}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P^* & \mathcal{C}_P^* & I \\ \mathcal{B}_P^* & \mathcal{D}_P^* & 0 \\ \mathcal{B}_u^* & \mathcal{D}_{zu}^* & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ \underline{w}(t) \\ \underline{u}_P(t) \end{bmatrix}.$$

We consider the controller and its adjoint

$$\begin{aligned} u(t) &= \mathcal{K}_P \mathbf{x}_P(t) + \mathcal{K}_\Psi \mathbf{x}_\Psi(t), \\ \begin{bmatrix} \underline{u}_P(t) \\ \underline{u}_\Psi(t) \end{bmatrix} &= \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Psi^* \end{bmatrix} \underline{y}(t). \end{aligned}$$

If $\Psi \neq I$, assume that $\mathbf{D}(\Psi)$ exists and that z, w are measurable. The primal filter, augmented with the state output $y_\Psi = \mathbf{x}_\Psi$, has realization

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{v}(t) \\ y_\Psi(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi & \mathcal{B}_\Psi \\ \mathcal{C}_\Psi & \mathcal{D}_\Psi \\ I & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ v(t) \end{bmatrix}.$$

Here $v := \begin{bmatrix} z \\ w \end{bmatrix}$ and $\tilde{v} := \begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix}$. Let $\hat{J} := \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$, with identity blocks of compatible dimensions, and assume that the static dual feedthrough $\mathbf{D}(\mathcal{D}_\Psi) = (\hat{J}^* \mathcal{D}_\Psi^* \hat{J})^{-1}$ exists. Then $\mathbf{D}(\Psi)$, including the additional input dual to y_Ψ , is

$$\begin{bmatrix} \mathcal{T}_\Psi^* \dot{\mathbf{x}}_\Psi(t) \\ \underline{\tilde{v}}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi & \mathcal{B}_\Psi & I \\ \mathcal{C}_\Psi & \mathcal{D}_\Psi & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \underline{v}(t) \\ \underline{u}_\Psi(t) \end{bmatrix},$$

where all dual-filter operators are expressed in the primal operators as

$$\begin{aligned} \underline{\mathcal{A}}_\Psi &:= \mathcal{A}_\Psi^* - \mathcal{C}_\Psi^* \hat{J} \mathbf{D}(\mathcal{D}_\Psi) \hat{J}^* \mathcal{B}_\Psi^*, \\ \underline{\mathcal{B}}_\Psi &:= \mathcal{C}_\Psi^* \hat{J} \mathbf{D}(\mathcal{D}_\Psi), \\ \underline{\mathcal{C}}_\Psi &:= -\mathbf{D}(\mathcal{D}_\Psi) \hat{J}^* \mathcal{B}_\Psi^*, \\ \underline{\mathcal{D}}_\Psi &:= \mathbf{D}(\mathcal{D}_\Psi). \end{aligned}$$

Here $\underline{v} := \begin{bmatrix} \underline{\tilde{z}} \\ \underline{\tilde{w}} \end{bmatrix}$ and $\underline{\tilde{v}} := \begin{bmatrix} \underline{\tilde{z}} \\ \underline{\tilde{w}} \end{bmatrix}$.

B. State Feedback LPI

Theorem 5. Let the realization of P satisfy Definition 7. Let $\Delta \in \Delta$, there exist dual hard IQC $(\Psi, \mathcal{M}_\Psi)$, such that,

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \Delta$.
- The IQC in Definition 10 is satisfied for all $\Delta \in \Delta$.
- The dual hard IQC $(\Psi, \mathcal{M}_\Psi)$ has a $(\mathcal{M}_{\mathbf{D}(\mathcal{U})} \mathbf{D}(\Psi), J_{\mathbf{n}_w, \mathbf{n}_z})$ factorization.

Moreover there exist, $\mathcal{Z}, \mathcal{P} \in \Pi_4$ with $\mathcal{P} = \mathcal{P}^* \succ 0$, such that the following statement holds

$$\begin{aligned} &\begin{bmatrix} \mathcal{T}^* \mathcal{P} \underline{\mathcal{A}}_0 + \mathcal{T}^* \mathcal{Z} \begin{bmatrix} \mathcal{B}_u^* & 0 \end{bmatrix} + (*) & \mathcal{T}^* \mathcal{P} \underline{\mathcal{B}}_0 + \mathcal{T}^* \mathcal{Z} \mathcal{D}_{zu}^* \\ (*) & \underline{\epsilon} I \end{bmatrix} \\ &+ \begin{bmatrix} \underline{\mathcal{C}}^* \\ \underline{\mathcal{D}}^* \end{bmatrix} \underline{\mathcal{Y}} \begin{bmatrix} \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \leq 0. \end{aligned} \quad (5)$$

where

$$\mathbf{D}(\Psi) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T} \dot{\underline{\xi}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}}_0 & \underline{\mathcal{B}}_0 \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{\xi}(t) \\ \underline{w}(t) \end{bmatrix} \right\} \quad (6)$$

and

$$\mathcal{Z} := \mathcal{P} \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Psi^* \end{bmatrix},$$

Then the interconnection $K \star P \star \Delta$ with $\mathbf{x}(0) = 0$ is robustly stable and robust performance is achieved.

C. $\Psi_\Delta = I$ controller realization

When $\Psi_\Delta = I$ and $\underline{\Psi}_\Delta = I$, no multiplier state is required in the controller. For a state-feedback operator $\mathcal{K}_P$, set $u(t) = \mathcal{K}_P \mathbf{x}_P(t)$. The resulting closed-loop PIE $\mathcal{K} \star \mathcal{P}$ is

$$\begin{bmatrix} \partial_t (\mathcal{T}_P \mathbf{x}_P)(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P + \mathcal{B}_u \mathcal{K}_P & \mathcal{B}_P \\ \mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P & \mathcal{D}_P \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ w(t) \end{bmatrix}.$$

D. Dynamic-filter controller realization

Following [14], assume that the uncertainty Δ is known and that z_Δ , $w_\Delta = \Delta z_\Delta$, and w_p are available to the controller. For $\underline{\Psi}_\Delta \neq I$, find the realization of $\Psi_\Delta = \mathbf{D}(\underline{\Psi}_\Delta)$,

$$\begin{bmatrix} \mathcal{T}_{\Psi_\Delta} \dot{\mathbf{x}}_{\Psi_\Delta}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_{\Psi_\Delta} & \mathcal{B}_{\Psi_\Delta}^z & \mathcal{B}_{\Psi_\Delta}^w \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\Psi_\Delta}(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}.$$

We impose the zero initial condition $\mathbf{x}_{\Psi_\Delta}(0) = 0$. The second block row gives $y(t) = \mathbf{x}_{\Psi_\Delta}(t)$. The complete controller has the PIE realization

$$\begin{bmatrix} \mathcal{T}_{\Psi_\Delta} \dot{\mathbf{x}}_{\Psi_\Delta}(t) \\ u(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_{\Psi_\Delta} & 0 & \mathcal{B}_{\Psi_\Delta}^z & \mathcal{B}_{\Psi_\Delta}^w \\ \mathcal{K}_\Psi & \mathcal{K}_P & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\Psi_\Delta}(t) \\ \mathbf{x}(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}.$$

Hence $\mathbf{x}_{\Psi_\Delta}$ is the controller state, $\mathbf{x}$, z_Δ , and w_Δ are its external inputs, and u is its output.

VI. NUMERICAL EXAMPLES

A. Spatial Lifting of IQC

We denote the extended L_2 space as $L_{2,e}$, let $\mathcal{F}$ denote the set of hard factorizations. We say the causal bounded uncertainty $\Delta : L_{2,e} \rightarrow L_{2,e}$ satisfies the hard iq

$$\left\langle p_T \Psi \begin{bmatrix} z \\ \Delta z \end{bmatrix}, p_T V \Psi \begin{bmatrix} z \\ \Delta z \end{bmatrix} \right\rangle_{L_2} \geq 0 \quad (7)$$

for all $T \geq 0$, $z \in L_{2,e}$, and hard factorizations $(\Psi, V) \in \mathcal{F}$, with $\Psi : L_{2,e}^n \rightarrow L_{2,e}^{\tilde{n}}$ and $V \in \mathbb{R}^{\tilde{n} \times \tilde{n}}$. Now we extend Δ pointwise over the spatial domain $[a, b]$. Denote the signal space $L_{\mathbf{H},e} = L_2([0, T) \times [a, b])$ for any signal $u \in L_{\mathbf{H},e}$

$$\langle u, u \rangle_{L_{\mathbf{H},e}} = \int_0^T \int_a^b |u(t, s)|^2 ds dt < \infty, \quad \forall T \geq 0 \quad (8)$$

Define the lifted uncertainty operation $(\hat{\Delta}v)(\cdot, s) = \Delta v(\cdot, s)$, i.e. Δ applied at almost every $s \in [a, b]$, the lifted hard factorization $(\hat{\Psi}, \hat{V})$, with $\hat{\Psi} : L_{\mathbf{H},e} \rightarrow L_{\mathbf{H},e}$ and $\hat{V} : L_2[a, b] \rightarrow L_2[a, b]$, such that $(\hat{\Psi}(\cdot, s), \hat{V}(\cdot, s)) \in \mathcal{F}$ for all $s \in [a, b]$.

Corollary 1. If (7) holds for all $T > 0$ and $z \in L_{2,e}[0, \infty)$, then

$$\left\langle p_T \hat{\Psi} \begin{bmatrix} u \\ \hat{\Delta} u \end{bmatrix}, p_T \hat{V} \hat{\Psi} \begin{bmatrix} u \\ \hat{\Delta} u \end{bmatrix} \right\rangle_{L_H} \geq 0 \quad (9)$$

for all $u \in L_{H,e}$ and $T \geq 0$

Proof. Assume (7) is satisfied. We need to show that (9) is also satisfied. First examine that

$$\left\langle p_T \hat{\Psi}(s) \begin{bmatrix} u(s) \\ \Delta u(s) \end{bmatrix}, p_T \hat{V}(s) \hat{\Psi}(s) \begin{bmatrix} u(s) \\ \Delta u(s) \end{bmatrix} \right\rangle_{L_2} \geq 0, \quad \forall s \in [a, b]$$

Thus by integrating over s we find

$$\begin{aligned} & \int_a^b \left\langle p_T \hat{\Psi}(s) \begin{bmatrix} u(s) \\ \Delta u(s) \end{bmatrix}, p_T \hat{V}(s) \hat{\Psi}(s) \begin{bmatrix} u(s) \\ \Delta u(s) \end{bmatrix} \right\rangle_{L_2} ds \\ &= \left\langle p_T \hat{\Psi} \begin{bmatrix} u \\ \Delta u \end{bmatrix}, p_T \hat{V} \hat{\Psi} \begin{bmatrix} u \\ \Delta u \end{bmatrix} \right\rangle_{L_H} \geq 0 \end{aligned}$$

■

Lemma 2 (Mixed-IQC). Let (Θ_i, J_{n_z, n_w}) denote a hard factorization, and $\gamma_i > 0$, then

$$\left\langle p_T \begin{bmatrix} P \\ I \end{bmatrix} w, p_T \sum_{i=1}^n \gamma_i^2 \Theta_i^* J \Theta_i \begin{bmatrix} P \\ I \end{bmatrix} w \right\rangle \leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2 \quad (10)$$

If and only if

$$\left\langle p_T \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w}, p_T \sum_{i=1}^n \frac{1}{\gamma_i^2} \mathbf{D}(\Theta_i)^* J \mathbf{D} \Theta_i \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} \right\rangle \leq -\epsilon \|p_T \underline{w}\|_{L_{[a,b]}}^2 \quad (11)$$

Proof. See Appendix A.6

■

Consider the nonlinear unstable pendulum system with dynamics

$$ml^2 \ddot{\theta}(t) + b\dot{\theta}(t) - mgl \sin(\theta(t)) = u(t), \quad (12)$$

where m, l, b, g, θ, u , denote the mass, length, damping coefficient, gravitational coefficient, angular position, and control input of the pendulum, respectively. After substituting $\phi(\theta) = \theta - \sin(\theta)$, $x_1 = \theta$, $x_2 = \dot{\theta}$, $z_\Delta = \theta$, $w_\Delta = \phi(\theta)$ we obtain

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ z_\Delta(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{g}{l} & -\frac{b}{ml^2} & -\frac{g}{l} & \frac{1}{ml^2} \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ w_\Delta(t) \\ u(t) \end{bmatrix}. \quad (13)$$

For $\theta \neq 0$, let $r(\theta) := \phi(\theta)/\theta = 1 - \sin(\theta)/\theta$, and define $r(0) := 0$ by continuity. The global bounds

$$-0.217234 \leq \frac{\sin(\theta)}{\theta} \leq 1, \quad \theta \neq 0,$$

imply that

$$0 \leq r(\theta) \leq 1.217234 = \beta, \quad \theta \in \mathbb{R}.$$

Hence, ϕ is globally sector bounded in $[\alpha, \beta] = [0, 1.217234]$ and satisfies

$$\phi(\theta)(1.217234\theta - \phi(\theta)) \geq 0, \quad \theta \in \mathbb{R}.$$

Moreover, $\phi(\theta)$ is also incrementally bounded or slope restricted such that

$$0 \leq \frac{\phi(x_1) - \phi(x_2)}{x_1 - x_2} \leq 2 = \hat{\beta}, \quad \forall x_1, x_2 \in \mathbb{R}$$

Since $\phi'(\theta) = 1 - \cos(\theta)$.

B. Dynamic Sector-Bounded IQC

In case of dynamic uncertainties for which we can find a sector bound we use the following IQC

$$\Pi_{\text{sec}} = \left\langle p_T \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix}, p_T \begin{bmatrix} \beta & -1 \\ 0 & 1 \end{bmatrix}^* \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix} \right\rangle \geq 0$$

C. Static Sector-Bounded IQC

For static sector bounded IQC we can use the Popov IQC

$$\Pi_{\text{Popov}} = \left\langle \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix}, \begin{bmatrix} 0 & -j\omega \\ j\omega & 0 \end{bmatrix} \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix} \right\rangle \geq 0 \quad (14)$$

In order to make this multiplier PN we need to have two conditions

- $\Pi_{\text{Popov}} \in \mathbb{RL}_\infty$ is solved by introducing a new channel $\hat{z}_\Delta = (s+1)z_\Delta$ and $\hat{\Delta} = \Delta \circ \frac{1}{s+1}$.
- $\Pi_{\text{Popov},11} > 0$ and $\Pi_{\text{Popov},22} < 0$ such that a hard factorization exists.

To obtain a strict PN multiplier after the loop transformation, we add a hard norm IQC directly on the loop-shifted channels. The validity of this IQC is established below. Using the assumption that $\theta(0) = 0$, the loop-shifted plant is

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \hat{z}_\Delta(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{g}{l} & -\frac{b}{ml^2} & -\frac{g}{l} & \frac{1}{ml^2} \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ w_\Delta(t) \\ u(t) \end{bmatrix}. \quad (15)$$

Set $w_\Delta := \hat{\Delta} \hat{z}_\Delta = \Delta z_\Delta$. Since $z_\Delta = (s+1)^{-1} \hat{z}_\Delta$, we have $\hat{z}_\Delta = z_\Delta + \dot{z}_\Delta$.

a) A hard norm IQC for the loop-shifted uncertainty:.

The sector condition implies $|w_\Delta(t)| \leq \beta |z_\Delta(t)|$ for almost every $t \geq 0$. For every $T \geq 0$, integration by parts gives

$$\begin{aligned} \|p_T \hat{z}_\Delta\|_{L_2}^2 &= \int_0^T |z_\Delta(t) + \dot{z}_\Delta(t)|^2 dt \\ &= \|p_T z_\Delta\|_{L_2}^2 + \|p_T \dot{z}_\Delta\|_{L_2}^2 + |z_\Delta(T)|^2 - |z_\Delta(0)|^2. \end{aligned}$$

Using $z_\Delta(0) = 0$ and the sector-induced norm bound, we obtain

$$\begin{aligned} & \|p_T \hat{z}_\Delta\|_{L_2}^2 - \frac{1}{\beta^2} \|p_T w_\Delta\|_{L_2}^2 \\ &= \left(\|p_T z_\Delta\|_{L_2}^2 - \frac{1}{\beta^2} \|p_T w_\Delta\|_{L_2}^2 \right) + \|p_T \dot{z}_\Delta\|_{L_2}^2 + |z_\Delta(T)|^2 \geq 0. \end{aligned}$$

Consequently, $\hat{\Delta}$ satisfies the hard IQC

$$\left\langle p_T \begin{bmatrix} \hat{z}_\Delta \\ \hat{\Delta} \hat{z}_\Delta \end{bmatrix}, p_T \Pi_\delta \begin{bmatrix} \hat{z}_\Delta \\ \hat{\Delta} \hat{z}_\Delta \end{bmatrix} \right\rangle_{L_2} \geq 0, \quad \Pi_\delta := \begin{bmatrix} I & 0 \\ 0 & -\beta^{-2} I \end{bmatrix}, \quad (16)$$

for every $T \geq 0$. This is an L_2 -gain IQC; it does not assert that $\hat{\Delta}$ has the same pointwise sector bound as Δ .

Define

$$L(j\omega) := \begin{bmatrix} (j\omega + 1)^{-1} & 0 \\ 0 & 1 \end{bmatrix}, \quad \hat{v}_\Delta := \begin{bmatrix} \hat{z}_\Delta \\ w_\Delta \end{bmatrix}.$$

The hard multiplier $\epsilon\Pi_\delta$ may now be added after the loop transformation. Define

$$\begin{aligned} \Pi_1(j\omega) &:= L(j\omega)^* \begin{bmatrix} 0 & \beta \\ \beta & -2 \end{bmatrix} L(j\omega) + \epsilon\Pi_\delta, \\ \Pi_2(j\omega) &:= L(j\omega)^* \begin{bmatrix} 0 & -j\omega \\ j\omega & 0 \end{bmatrix} L(j\omega) + \epsilon\Pi_\delta. \end{aligned}$$

D. Slope restricted IQC

Conditions for Zames-Falb multiplier are Δ is a monotone increasing uncertainty, let $M = I - H$ with $\|H\|_1 \leq 1$, and $h(t) \geq 0$ for non-odd functions.

$$\Pi_3(j\omega) := L(j\omega)^* \begin{bmatrix} 0 & \hat{\beta}M(j\omega)^* \\ \hat{\beta}M(j\omega) & -(M(j\omega) + M(j\omega)^*) \end{bmatrix} L(j\omega) + \epsilon\Pi_\delta.$$

Thus, Π_1 , Π_2 , and Π_3 all have the uniform upper-left block ϵI . Their lower-right blocks are strictly negative provided $M(j\omega) + M(j\omega)^* \succeq 0$ for every ω . A common realization is $M = I - \prod_{i=1}^n \frac{k_i}{s+a_i}$ with $a_i > 0$, $(k_i > 0 \implies h(t) > 0)$, and $\prod_{i=1}^n \frac{a_i}{k_i} \leq 1$ [12].

VII. CONCLUSION

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APPENDIX A MATHEMATICAL PROOFS

A. Proof of Theorem 1

Suppose that (2) holds. This implies,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} + \left\langle p_T \Psi_P \begin{bmatrix} P_{p\Delta} \\ 0 \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_P} \Psi_P \begin{bmatrix} P_{p\Delta} \\ 0 \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2.$$

By Definition 11 we can then infer,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2.$$

Then, using Theorem 6 in [11], we find that **robust stability** is guaranteed. Moreover, rewriting (2) as,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} + \left\langle p_T \Psi_P \begin{bmatrix} z_P \\ w_P \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_P} \Psi_P \begin{bmatrix} z_P \\ w_P \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w_P\|_{\mathbf{L}_{[a,b]}}^2),$$

and given that Ψ_Δ and $\mathcal{M}_{\mathcal{V}_\Delta}$ satisfy the IQC in Definition 10 leads to,

$$\left\langle p_T \Psi_P \begin{bmatrix} z_P \\ w_P \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_P} \Psi_P \begin{bmatrix} z_P \\ w_P \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_P\|_{\mathbf{L}_{[a,b]}}^2.$$

Thus by Definition 11 **robust performance** is achieved.

B. Proof of Lemma 1

We prove the primal and dual identities in block form. Set

$$v := \begin{bmatrix} z \\ w \end{bmatrix}, \quad \tilde{v} := \begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix}, \quad v^\dagger := \begin{bmatrix} z \\ \tilde{w} \end{bmatrix}, \quad \tilde{v}^\dagger := \begin{bmatrix} \tilde{z} \\ w \end{bmatrix},$$

Let

$$R_{zw} := \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}, \quad R_{wz} := \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix},$$

where the block dimensions are determined by the indicated channel order, so that

$$R_{zw} \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} w \\ z \end{bmatrix}, \quad R_{wz} \begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} z \\ w \end{bmatrix}.$$

We write R when the channel order is clear from context. Realize the plant and filter as

$$P := \left\{ \begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P \\ z \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P & \mathcal{B}_P \\ \mathcal{C}_P & \mathcal{D}_P \end{bmatrix} \begin{bmatrix} \mathbf{x}_P \\ w \end{bmatrix}, \quad \Theta := \left\{ \begin{bmatrix} \mathcal{T} \dot{\mathbf{x}} \\ \tilde{v} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ v \end{bmatrix} \right. \\ \left. \mathcal{B} := \begin{bmatrix} \mathcal{B}_1 & \mathcal{B}_2 \end{bmatrix}, \quad \mathcal{C} := \begin{bmatrix} \mathcal{C}_1 \\ \mathcal{C}_2 \end{bmatrix}, \quad \mathcal{D} := \begin{bmatrix} \mathcal{D}_{11} & \mathcal{D}_{12} \\ \mathcal{D}_{21} & \mathcal{D}_{22} \end{bmatrix} \right.$$

Primal. The exchange operator places the external signal first:

$$\Theta^\dagger R \begin{bmatrix} \tilde{w} \\ z \end{bmatrix} = \begin{bmatrix} \tilde{z} \\ w \end{bmatrix}.$$

In the original port order,

$$\Theta^\dagger := \left\{ \begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ \tilde{v}^\dagger(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^\dagger & \mathcal{B}^\dagger \\ \mathcal{C}^\dagger & \mathcal{D}^\dagger \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ v^\dagger(t) \end{bmatrix} \right\}, \quad (17)$$

The corresponding realization operators are

$$\begin{bmatrix} \mathcal{A}^\dagger & \mathcal{B}^\dagger \\ \mathcal{C}^\dagger & \mathcal{D}^\dagger \end{bmatrix} := \begin{bmatrix} \mathcal{A} & \mathcal{B}_1 & 0 \\ \mathcal{C}_1 & \mathcal{D}_{11} & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} \mathcal{B}_2 \\ \mathcal{D}_{12} \\ I \end{bmatrix} \mathcal{D}_{22}^{-1} \begin{bmatrix} -\mathcal{C}_2 & -\mathcal{D}_{21} & I \end{bmatrix}. \quad (18)$$

Substituting $z = \mathcal{C}_P \mathbf{x}_P + \mathcal{D}_P w$ into the last row gives

$$w = \Lambda (\mathcal{D}_{22}^\dagger)^{-1} \left(\mathcal{D}_{21}^\dagger \mathcal{C}_P \mathbf{x}_P + \mathcal{C}_2^\dagger \mathbf{x} + \mathcal{D}_{22}^\dagger \tilde{w} \right), \quad (19)$$

and therefore

$$z = \mathcal{C}_P \mathbf{x}_P + \mathcal{D}_P \Lambda (\mathcal{D}_{22}^\dagger)^{-1} \left(\mathcal{D}_{21}^\dagger \mathcal{C}_P \mathbf{x}_P + \mathcal{C}_2^\dagger \mathbf{x} + \mathcal{D}_{22}^\dagger \tilde{w} \right). \quad (20)$$

Substitution of (19)–(20) into the remaining rows gives

$$\begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P \\ \mathcal{T} \dot{\mathbf{x}} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P & 0 & 0 \\ \mathcal{B}_1^\dagger \mathcal{C}_P & \mathcal{A}^\dagger & \mathcal{B}_2^\dagger \\ \mathcal{D}_{11}^\dagger \mathcal{C}_P & \mathcal{C}_1^\dagger & \mathcal{D}_{12}^\dagger \\ 0 & 0 & I \end{bmatrix} + \begin{bmatrix} \mathcal{B}_P \\ \mathcal{B}_1^\dagger \mathcal{D}_P \\ \mathcal{D}_{11}^\dagger \mathcal{D}_P \\ 0 \end{bmatrix} \Lambda (\mathcal{D}_{22}^\dagger)^{-1} \begin{bmatrix} \mathcal{D}_{21}^\dagger \mathcal{C}_P & \mathcal{C}_2^\dagger & \mathcal{D}_{22}^\dagger \end{bmatrix} \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x} \\ \tilde{w} \end{bmatrix} \quad (21)$$

Here

$$\Lambda := (\mathcal{D}_{21} \mathcal{D}_P + \mathcal{D}_{22})^{-1}, \quad \Lambda (\mathcal{D}_{22}^\dagger)^{-1} = (I - \mathcal{D}_{21}^\dagger \mathcal{D}_P)^{-1}. \quad (22)$$

Define the coordinate changes by

$$\begin{bmatrix} \mathbf{x}_P \\ \mathbf{x} \\ \tilde{w} \end{bmatrix} = \underbrace{\begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_{21} \mathcal{C}_P & \mathcal{C}_2 & \Lambda^{-1} \end{bmatrix}}_{\Gamma} \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x} \\ w \end{bmatrix}, \quad \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x} \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\Lambda \mathcal{D}_{21} \mathcal{C}_P & -\Lambda \mathcal{C}_2 & \Lambda \end{bmatrix}}_{\Gamma^{-1}} \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x} \\ \tilde{w} \end{bmatrix}.$$

Using (18) and (22), right multiplication of (21) by Γ gives

$$\begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P \\ \mathcal{T} \dot{\mathbf{x}} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P & 0 & \mathcal{B}_P \\ \mathcal{B}_1^\dagger \mathcal{C}_P & \mathcal{A}^\dagger & \mathcal{B}_1^\dagger \mathcal{D}_P \\ \mathcal{D}_{11}^\dagger \mathcal{C}_P & \mathcal{C}_1^\dagger & \mathcal{D}_{11}^\dagger \mathcal{D}_P \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \mathcal{B}_2^\dagger \\ \mathcal{D}_{12}^\dagger \\ I \end{bmatrix} (\mathcal{D}_{22}^\dagger)^{-1} \begin{bmatrix} -\mathcal{D}_{21}^\dagger \mathcal{C}_P & -\mathcal{C}_2^\dagger & I - \mathcal{D}_{21}^\dagger \mathcal{D}_P \end{bmatrix} \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x} \\ w \end{bmatrix}. \quad (23)$$

$(\Theta^\dagger)^\dagger = \Theta.$

Consequently, the lower three rows of (23) reduce to

$$\begin{bmatrix} \mathcal{T} \dot{\mathbf{x}} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} 0 & I & 0 \\ \mathcal{C}_P & 0 & \mathcal{D}_P \\ 0 & 0 & I \end{bmatrix} \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x} \\ w \end{bmatrix}, \quad (24)$$

which, together with the plant row, gives

$$\begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P \\ \mathcal{T} \dot{\mathbf{x}} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P & 0 & \mathcal{B}_P \\ \mathcal{B}_1 \mathcal{C}_P & \mathcal{A} & \mathcal{B}_1 \mathcal{D}_P + \mathcal{B}_2 \\ \mathcal{D}_{11} \mathcal{C}_P & \mathcal{C}_1 & \mathcal{D}_{11} \mathcal{D}_P + \mathcal{D}_{12} \\ \mathcal{D}_{21} \mathcal{C}_P & \mathcal{C}_2 & \mathcal{D}_{21} \mathcal{D}_P + \mathcal{D}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x} \\ w \end{bmatrix}.$$

$\Theta \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} P \star (\Theta^\dagger R) \\ I \end{bmatrix} \mathcal{M}_\Gamma w.$

Dual. Set

$$\underline{v} := \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix}, \quad \tilde{\underline{v}} := \begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix}, \quad \hat{J} := \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}.$$

The dual filter satisfies

$$\mathbf{D}(\Theta) = (\hat{J}^* \Theta^\top \hat{J})^{-1}$$

and has realization

$$\begin{bmatrix} \mathcal{T}^* \dot{\underline{x}} \\ \underline{\tilde{v}} \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{x} \\ \underline{v} \end{bmatrix},$$

with

$$\begin{aligned} \underline{\mathcal{A}} &:= \mathcal{A}^* - \mathcal{C}^* \hat{J} \mathbf{D}(\mathcal{D}) \hat{J}^* \mathcal{B}^*, & \underline{\mathcal{B}} &:= \mathcal{C}^* \hat{J} \mathbf{D}(\mathcal{D}), \\ \underline{\mathcal{C}} &:= -\mathbf{D}(\mathcal{D}) \hat{J}^* \mathcal{B}^*, & \underline{\mathcal{D}} &:= \mathbf{D}(\mathcal{D}) = (\hat{J}^* \mathcal{D}^* \hat{J})^{-1}, \end{aligned} \quad (25)$$

Block inversion and (18) yield

$$\begin{aligned} \mathbf{D}(\mathcal{D}) &= \begin{bmatrix} (\mathcal{D}_{22}^\dagger)^* - (\mathcal{D}_{12}^\dagger)^* \left((\mathcal{D}_{11}^\dagger)^* \right)^{-1} (\mathcal{D}_{21}^\dagger)^* & (\mathcal{D}_{12}^\dagger)^* \left((\mathcal{D}_{11}^\dagger)^* \right)^{-1} \\ - \left((\mathcal{D}_{11}^\dagger)^* \right)^{-1} (\mathcal{D}_{21}^\dagger)^* & \left((\mathcal{D}_{11}^\dagger)^* \right)^{-1} \end{bmatrix} \\ &= \begin{bmatrix} (\mathcal{D}_{22}^\dagger)^* & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} (\mathcal{D}_{12}^\dagger)^* \\ I \end{bmatrix} \left((\mathcal{D}_{11}^\dagger)^* \right)^{-1} \begin{bmatrix} -(\mathcal{D}_{21}^\dagger)^* & I \end{bmatrix}. \end{aligned}$$

Thus (25) becomes

$$\begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}}_1 & \underline{\mathcal{B}}_2 \\ \underline{\mathcal{C}}_1 & \underline{\mathcal{D}}_{11} & \underline{\mathcal{D}}_{12} \\ \underline{\mathcal{C}}_2 & \underline{\mathcal{D}}_{21} & \underline{\mathcal{D}}_{22} \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} -\mathcal{C}_2^* & \mathcal{C}_1^* \\ I & 0 \\ 0 & I \end{bmatrix} \mathbf{D}(\mathcal{D}) \begin{bmatrix} \mathcal{B}_2^* & I & 0 \\ -\mathcal{B}_1^* & 0 & I \end{bmatrix}.$$

Substitution gives

$$\begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}}_1 & \underline{\mathcal{B}}_2 \\ \underline{\mathcal{C}}_1 & \underline{\mathcal{D}}_{11} & \underline{\mathcal{D}}_{12} \\ \underline{\mathcal{C}}_2 & \underline{\mathcal{D}}_{21} & \underline{\mathcal{D}}_{22} \end{bmatrix} := \begin{bmatrix} (\mathcal{A}^\dagger)^* & (\mathcal{C}_2^\dagger)^* & 0 \\ (\mathcal{B}_2^\dagger)^* & (\mathcal{D}_{22}^\dagger)^* & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} (\mathcal{C}_1^\dagger)^* \\ (\mathcal{D}_{12}^\dagger)^* \\ I \end{bmatrix} \left((\mathcal{D}_{11}^\dagger)^* \right)^{-1} \begin{bmatrix} -(\mathcal{B}_1^\dagger)^* & -(\mathcal{D}_{21}^\dagger)^* & I \end{bmatrix}. \quad (26)$$

The full block realization of $R(\Theta^\dagger)^\top R$ is

$$\begin{bmatrix} \mathcal{T}^* \underline{\mathbf{x}} \\ \underline{\tilde{z}} \\ \underline{w} \end{bmatrix} = \begin{bmatrix} (\mathcal{A}^\dagger)^* & (\mathcal{C}_2^\dagger)^* & (\mathcal{C}_1^\dagger)^* \\ (\mathcal{B}_2^\dagger)^* & (\mathcal{D}_{22}^\dagger)^* & (\mathcal{D}_{12}^\dagger)^* \\ (\mathcal{B}_1^\dagger)^* & (\mathcal{D}_{21}^\dagger)^* & (\mathcal{D}_{11}^\dagger)^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}} \\ \underline{\tilde{z}} \\ \underline{\tilde{w}} \end{bmatrix}.$$

Comparison with (18) and (26) gives

$$\mathbf{D}(\Theta) = (R(\Theta^\dagger)^\top R)^\dagger, \quad \mathbf{D}(\Theta)^\dagger = R(\Theta^\dagger)^\top R.$$

The dual signals therefore satisfy

$$\begin{bmatrix} \underline{\tilde{z}} \\ \underline{w} \end{bmatrix} = \mathbf{D}(\Theta)^\dagger R \begin{bmatrix} \underline{\tilde{w}} \\ \underline{z} \end{bmatrix}, \quad \underline{z} = P^\top \underline{w}.$$

Using $RR = I$,

$$\mathbf{D}(\Theta)^\dagger R = R(\Theta^\dagger)^\top.$$

Transposition of the primal identity gives

$$P^\top \star (R(\Theta^\dagger)^\top) = (P \star (\Theta^\dagger R))^\top.$$

Hence

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} P^\top \star (R(\Theta^\dagger)^\top) \\ I \end{bmatrix} \mathcal{M}_{\underline{\Gamma}} \underline{w}.$$

Finally, using (26), define the dual interconnection operator in terms of the primal realization operators as

$$\underline{\Lambda} := \left(I - (\mathcal{D}_{21}^\dagger)^* \mathcal{D}_P^* \right)^{-1} (\mathcal{D}_{11}^\dagger)^*,$$

with coordinate changes

$$\begin{aligned} \begin{bmatrix} \underline{\mathbf{x}}_P \\ \underline{\mathbf{x}} \\ \underline{\tilde{w}} \end{bmatrix} &= \underbrace{\begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_{21} \mathcal{B}_P^* & \mathcal{C}_2 & \underline{\Lambda}^{-1} \end{bmatrix}}_{\underline{\Gamma}} \begin{bmatrix} \underline{\mathbf{x}}_P \\ \underline{\mathbf{x}} \\ \underline{w} \end{bmatrix}, \\ \begin{bmatrix} \underline{\mathbf{x}}_P \\ \underline{\mathbf{x}} \\ \underline{w} \end{bmatrix} &= \underbrace{\begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\underline{\Lambda} \mathcal{D}_{21} \mathcal{B}_P^* & -\underline{\Lambda} \mathcal{C}_2 & \underline{\Lambda} \end{bmatrix}}_{\underline{\Gamma}^{-1}} \begin{bmatrix} \underline{\mathbf{x}}_P \\ \underline{\mathbf{x}} \\ \underline{\tilde{w}} \end{bmatrix}. \end{aligned}$$

C. Proof of Theorem 3

Assume that the primal statement of the theorem holds. Then there exists a sufficiently small $\epsilon > 0$ such that

$$\left\langle p_T \Theta \begin{bmatrix} P \\ I \end{bmatrix} w, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} P \\ I \end{bmatrix} w \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2$$

for all $w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}$ and $T \geq 0$. By lemma 1, the system $\Theta \begin{bmatrix} P \\ I \end{bmatrix}$ has a realization of the form

$$\left\langle p_T \begin{bmatrix} P \star (\Theta^\dagger R) \\ I \end{bmatrix} \Gamma w, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \begin{bmatrix} P \star (\Theta^\dagger R) \\ I \end{bmatrix} \Gamma w \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2$$

substituting $\tilde{w} = \Gamma w$, $\Gamma^{-1}\tilde{w} = w$, gives

$$\left\langle p_T \begin{bmatrix} P \star (\Theta^\dagger R) \\ I \end{bmatrix} \tilde{w}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \begin{bmatrix} P \star (\Theta^\dagger R) \\ I \end{bmatrix} \tilde{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T \Gamma^{-1} \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2$$

Since Γ is causal and bounded, $\|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}} \leq \|\Gamma\|_{\mathcal{L}(\mathbf{L}_{[a,b]})} \|p_T \Gamma^{-1} \tilde{w}\|_{\mathbf{L}_{[a,b]}}$, and therefore

$$\|p_T (P \star (\Theta^\dagger R)) \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2 \leq (1 - \frac{\epsilon}{\|\Gamma\|_{\mathcal{L}(\mathbf{L}_{[a,b]})}^2}) \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2.$$

Then by Theorem 2 and

$$(P \star (\Theta^\dagger R))^\top = P^\top \star (R(\Theta^\dagger)^\top),$$

we find

$$\|p_T (P^\top \star (R(\Theta^\dagger)^\top)) \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2 \leq (1 - \frac{\epsilon}{\|\Gamma\|_{\mathcal{L}(\mathbf{L}_{[a,b]})}^2}) \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2$$

recovering the dual statement of the theorem

$$\left\langle p_T \begin{bmatrix} P^\top \star (R(\Theta^\dagger)^\top) \\ I \end{bmatrix} \tilde{w}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \begin{bmatrix} P^\top \star (R(\Theta^\dagger)^\top) \\ I \end{bmatrix} \tilde{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\frac{\epsilon}{\|\Gamma\|_{\mathcal{L}(\mathbf{L}_{[a,b]})}^2} \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2$$

then $\tilde{w} = \Gamma w$, $\Gamma^{-1}\tilde{w} = w$ gives

$$\left\langle p_T \begin{bmatrix} P^\top \star (R(\Theta^\dagger)^\top) \\ I \end{bmatrix} \Gamma w, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \begin{bmatrix} P^\top \star (R(\Theta^\dagger)^\top) \\ I \end{bmatrix} \Gamma w \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\frac{\epsilon}{\|\Gamma\|_{\mathcal{L}(\mathbf{L}_{[a,b]})}^2} \|p_T \Gamma w\|_{\mathbf{L}_{[a,b]}}^2$$

Since Γ^{-1} is causal and bounded, $\|p_T w\|_{\mathbf{L}_{[a,b]}} \leq \|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L}_{[a,b]})} \|p_T \Gamma w\|_{\mathbf{L}_{[a,b]}}$, and this gives the dual statement

$$\left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} w, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} w \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2$$

where

$$\underline{\epsilon} := \frac{\epsilon}{\|\Gamma\|_{\mathcal{L}(\mathbf{L}_{[a,b]})}^2 \|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L}_{[a,b]})}^2} > 0.$$

Conversely, starting from the dual statement and reversing the same argument gives the corresponding primal margin

$$\epsilon := \frac{\underline{\epsilon}}{\|\Gamma\|_{\mathcal{L}(\mathbf{L}_{[a,b]})}^2 \|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L}_{[a,b]})}^2} > 0.$$

D. Proof of Theorem 4

If the primal LPI holds, the standard KYP integration gives the hard IQC

$$\left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_\Psi \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2.$$

Theorem 1 then gives the result. It remains to consider the dual LPI. Define $V(\mathbf{x}) = \langle \mathcal{T}\mathbf{x}, \mathcal{P}\mathcal{T}\mathbf{x} \rangle_{\mathbf{L}_{[a,b]}} \geq 0$. For any $\underline{w} \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}$, suppose $\underline{v}$ satisfies (4) for some $\underline{x} \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_x}$. By (4), we have

$$\left\langle \begin{bmatrix} \underline{x}(t) \\ \underline{w}(t) \end{bmatrix}, \begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \underline{\epsilon} I \end{bmatrix} \begin{bmatrix} \underline{x}(t) \\ \underline{w}(t) \end{bmatrix} \right\rangle + \langle \underline{v}(t), \mathcal{V} \underline{v}(t) \rangle \leq 0.$$

After integrating in time, we obtain

$$\underline{\epsilon} \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2 + \langle p_T \underline{v}, p_T \mathcal{V} \underline{v} \rangle_{\mathbf{L}_{[a,b]}} \leq -V(\mathbf{x}(T)) + V(\mathbf{x}(0)).$$

Since $\mathbf{x}(0) = 0$ and $V(\mathbf{x}(T)) \geq 0$,

$$\left\langle p_T \underline{\Psi} \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix}, p_T \mathcal{M}_\Psi \underline{\Psi} \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\underline{\epsilon} \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2.$$

Using the $(\mathbf{D}(\Theta), J_{\mathbf{n}_w, \mathbf{n}_z})$ factorization, this is

$$\left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\Theta) \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\underline{\epsilon} \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2.$$

By Theorem 3, there exists $\epsilon > 0$ such that

$$\left\langle p_T \Theta \begin{bmatrix} z \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2.$$

Using the $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ factorization gives the primal hard IQC, and Theorem 1 gives robust stability and robust performance.

E. Proof of Theorem 5

Since $\mathcal{P} \succ 0$, define $\mathcal{K}_P$ and $\mathcal{K}_\Psi$ by

$$\mathcal{Z}^* = [\mathcal{K}_P \quad \mathcal{K}_\Psi] \mathcal{P}.$$

Set

$$\underline{\mathcal{A}} := \underline{\mathcal{A}}_0 + \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Psi^* \end{bmatrix} \begin{bmatrix} \mathcal{B}_u^* & 0 \end{bmatrix}, \quad \underline{\mathcal{B}} := \underline{\mathcal{B}}_0 + \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Psi^* \end{bmatrix} \mathcal{D}_{zu}^*.$$

Then (5) is the dual KYP inequality for

$$\mathbf{D}(\Psi) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix}, \quad P_K := K \star P, \quad \underline{\mathbf{x}} := \begin{bmatrix} \mathbf{x}_P \\ \mathbf{x}_\Psi \end{bmatrix} \right\}.$$

For

$$V(\underline{\mathbf{x}}) := \langle \mathcal{T}\underline{\mathbf{x}}, \mathcal{P}\mathcal{T}\underline{\mathbf{x}} \rangle \geq 0,$$

the standard storage argument gives, for zero initial conditions,

$$\left\langle p_T \mathbf{D}(\Psi) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w}, p_T \mathcal{M}_{\underline{\mathcal{V}}} \mathbf{D}(\Psi) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2.$$

Define

$$\Theta := \mathcal{M}_{\mathcal{U}} \Psi.$$

Using

$$\underline{\mathcal{V}} = \mathbf{D}(\mathcal{U})^* J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\mathcal{U}), \quad \mathbf{D}(\Theta) = \mathcal{M}_{\mathbf{D}(\mathcal{U})} \mathbf{D}(\Psi),$$

the preceding inequality becomes

$$\left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\Theta) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2. \quad (27)$$

By Lemma 1, there exist invertible coordinate operators $\mathcal{M}_{\Gamma_K}$ and $\mathcal{M}_{\Gamma_K}$ such that

$$\begin{aligned} \Theta \begin{bmatrix} P_K \\ I \end{bmatrix} w &= \begin{bmatrix} P_K \star (\Theta^\dagger R) \\ I \end{bmatrix} \mathcal{M}_{\Gamma_K} w, \\ \mathbf{D}(\Theta) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w} &= \begin{bmatrix} P_K^\top \star (R(\Theta^\dagger)^\top) \\ I \end{bmatrix} \mathcal{M}_{\Gamma_K} \underline{w}, \end{aligned}$$

where

$$\mathbf{D}(\Theta)^\dagger R = R(\Theta^\dagger)^\top, \quad P_K^\top \star (R(\Theta^\dagger)^\top) = (P_K \star (\Theta^\dagger R))^\top.$$

Thus (27) is the dual feasibility condition associated with the primal filtered system. Theorem 3 therefore gives some $\epsilon > 0$ such that

$$\left\langle p_T \Theta \begin{bmatrix} P_K \\ I \end{bmatrix} w, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} P_K \\ I \end{bmatrix} w \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2.$$

Since $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ is the primal hard-IQC factorization, Theorem 1 gives robust stability and robust performance of $K \star P \star \Delta$.

F. Proof of Lemma 2

To show sufficiency, assume (10) is satisfied. We need to show that (11) is also satisfied. Factorize (10) as

$$\left\langle p_T \begin{bmatrix} P \\ I \end{bmatrix} w, p_T \begin{bmatrix} \gamma_1 I \\ \vdots \\ \gamma_n I \end{bmatrix}^* \begin{bmatrix} \Theta_1^* J \Theta_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \Theta_n^* J \Theta_n \end{bmatrix} \begin{bmatrix} \gamma_1 I \\ \vdots \\ \gamma_n I \end{bmatrix} \begin{bmatrix} P \\ I \end{bmatrix} w \right\rangle \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2 \quad (28)$$

Define $\hat{z} = [\gamma_1 I \quad \cdots \quad \gamma_n I]^* z$, $\hat{w} = [\gamma_1 I \quad \cdots \quad \gamma_n I]^* w$, then $[\frac{1}{\gamma_1} I \quad \cdots \quad \frac{1}{\gamma_n} I] \hat{w} = w$. Then we can reorganize (28) to show

$$\left\langle p_T \Theta \begin{bmatrix} \hat{P} \\ I \end{bmatrix} \hat{w}, p_T J \Theta \begin{bmatrix} \hat{P} \\ I \end{bmatrix} \hat{w} \right\rangle \leq -\epsilon \|p_T [\frac{1}{\gamma_1} I \quad \cdots \quad \frac{1}{\gamma_n} I] \hat{w}\|_{\mathbf{L}_{[a,b]}}^2$$

where,

$$\hat{P} \hat{w} := \left\{ \begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ \hat{z} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} [\frac{1}{\gamma_1} I \quad \cdots \quad \frac{1}{\gamma_n} I] \\ [\gamma_1 I \quad \cdots \quad \gamma_n I]^* \mathcal{C} & [\gamma_1 I \quad \cdots \quad \gamma_n I]^* \mathcal{D} [\frac{1}{\gamma_1} I \quad \cdots \quad \frac{1}{\gamma_n} I] \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \hat{w}(t) \end{bmatrix} \right\}$$

then,

$$\hat{P}^\top \underline{\hat{w}} := \left\{ \begin{bmatrix} \mathcal{T} \dot{\underline{\mathbf{x}}}(t) \\ \underline{\hat{z}} \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & & \\ \left[\frac{1}{\gamma_1} I & \cdots & \frac{1}{\gamma_n} \right]^* \mathcal{B}^* & \left[\frac{1}{\gamma_1} I & \cdots & \frac{1}{\gamma_n} \right]^* \mathcal{D}^* \begin{bmatrix} \mathcal{C}^* \begin{bmatrix} \gamma_1 I & \cdots & \gamma_n I \end{bmatrix} \\ \gamma_1 I & \cdots & \gamma_n I \end{bmatrix} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{\hat{w}}(t) \end{bmatrix} \right.$$

and by Theorem 3

$$\left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} \hat{P}^\top \\ I \end{bmatrix} \underline{\hat{w}}, p_T J \mathbf{D}(\Theta) \begin{bmatrix} \hat{P}^\top \\ I \end{bmatrix} \underline{\hat{w}} \right\rangle \leq -\epsilon \|p_T \begin{bmatrix} \frac{1}{\gamma_1} I & \cdots & \frac{1}{\gamma_n} \end{bmatrix} \underline{\hat{w}}\|_{\mathbf{L}_{[a,b]}}^2$$